

When Does a Stable Winner Exist? Coherence in Linear Gaussian Dynamical Systems with an Evolution Export Test*

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Abstract

When comparing descriptions of a dynamical system across multiple prediction tasks, it is natural to ask whether one description is globally best — a stable winner over the full target family. We give an exact answer for finite-dimensional linear Gaussian dynamical systems (LGDS): a stable winner exists within the rank- r observer class if and only if the target family is r -coherent, meaning that all target information matrices share a common top- r eigenspace. When coherence fails, no rank- r observer is simultaneously optimal for all targets and the global-privilege gap is strictly positive. We then extract an empirical diagnostic from the theorem: supported pairwise sign flips across a target family diagnose empirical incoherence and motivate a prediction of winner-absence. We test this diagnostic in an individual-based evolutionary simulation battery spanning six regimes, four descriptions, and two task families (predictive and causal). In a pre-aggregated adapter analysis, all 11 classified incoherent regime-family cells are winner-absent (0 mismatches; pre-registered pass condition met). Current support is strongest for the negative half of the diagnostic: empirically incoherent cells are winner-absent. No coherent cells were observed, so the positive half remains untested.

1 Introduction

Science frequently compares descriptions of the same system. A coarse spatial summary and a fine structural count are both legitimate measurements of a cellular automaton; a gene-frequency composition and a lineage-level profile are both legitimate descriptions of an evolving population. When predictions are made from each description, they may agree or disagree, and one description may be more accurate than another for a particular future quantity.

The natural next question is whether one description is *globally* best: not merely better for one target, but the best description to use regardless of what is being predicted. In practice this question is often answered implicitly, by convention or by computational convenience. The formal question — when is a stable winner over a target family even possible? — has not, to our knowledge, been given a precise answer in this form.

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This paper gives one. For finite-dimensional linear Gaussian dynamical systems, among rank- r linear observers and squared-error prediction targets, a stable winner exists if and only if the target family satisfies an algebraic condition we call r -coherence: all target information matrices share a common top- r eigenspace. When this condition fails, the loss from any fixed observer applied to a non-native target is strictly positive, and no observer can be simultaneously optimal for every target in the family.

The theorem is narrow by design. It does not claim that description-dependent privilege is universal, that all real systems are incoherent, or that any particular prior empirical result is an instance of the theorem. It claims only that, in the exact LGDS setting, the coherence condition is both necessary and sufficient for the existence of a stable winner.

The theorem’s operative structure — pairwise sign stability across a target family — suggests an empirical diagnostic that may travel beyond LGDS. We formalise this diagnostic (Section 4) and test it in an evolutionary simulation battery (Section 6). The test is not a theorem extension; it is an empirical probe of transportability.

Four prior empirical results motivated the formal question. In post-transient Game of Life, the scale that best predicts a future observable depends on which observable is chosen [Bhatia, 2026b]. In kinetic Ising dynamics, fine-description advantage reverses between ordered and disordered phases [Bhatia, 2026d]. In an open Bose–Hubbard chain, a causal handle effective in one coupling regime reverses in another [Bhatia, 2026a]. In three-dimensional N-body simulations, the winning description depends on the initial-condition family [Bhatia, 2026f]. Each result posed a winner-stability question in a different system; none supplied a formal criterion for when a stable winner should exist. Section 5 repositions these results as empirical pressure for the question the theorem now answers, not as theorem-governed instances.

The paper is organised as follows. Section 2 defines descriptions, target families, scores, coherence, and stable winnerhood. Section 3 states and proves the main theorem. Section 4 extracts the exported diagnostic. Section 5 briefly repositions the prior empirical results. Section 6 reports the evolution export test. Section 7 states scope and limitations. Section 8 discusses implications.

2 Framework

2.1 Descriptions and target families

Let $\mathcal{X}$ denote the state space of a dynamical system. A *description* is a measurable function $D : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps a system state to a fixed-dimensional summary. We work with a finite family of descriptions $\mathcal{D} = \{D_1, \dots, D_k\}$, where each D_i represents a distinct representational choice — a coarse-graining, an observable class, or a level of description.

A *target family* is a finite set $\mathcal{T} = \{Y_1, \dots, Y_m\}$ of scalar prediction targets, where each Y_j is a measurable function of the system state at a fixed future horizon $\tau \geq 1$. The target family formalises the question “what is the description for?”

2.2 Scores

For each description–target pair (D_i, Y_j) , a *score* $s(D_i; Y_j) \in \mathbb{R}$ measures predictive performance, with higher values indicating better performance. The score is computed on held-out

data or, where exact, from an analytic risk formula. The pairwise gap

$$\Delta_{ij}(Y) = s(D_i; Y) - s(D_j; Y) \quad (1)$$

is positive when D_i outperforms D_j on target Y .

A target $Y \in \mathcal{T}$ is *estimable* for the pair (D_i, D_j) if the score difference $\Delta_{ij}(Y)$ can be evaluated (in the empirical setting, if the sample size exceeds a pre-specified floor). A gap is *supported* if the estimable gap is nonzero — either exactly, in the analytic case, or with a confidence interval that excludes zero, in the empirical case.

2.3 Coherence

Definition 2.1 (Coherence). *A target family $\mathcal{T}$ is coherent with respect to description family $\mathcal{D}$ if, for every ordered pair (D_i, D_j) with $i \neq j$, the sign of $\Delta_{ij}(Y)$ is the same across all estimable $Y \in \mathcal{T}$ where the gap is supported.*

Equivalently, no pair (D_i, D_j) exhibits a sign flip: two estimable targets $Y_a, Y_b \in \mathcal{T}$ such that $\Delta_{ij}(Y_a) > 0$ and $\Delta_{ij}(Y_b) < 0$ with both gaps supported.

Coherence is a property of the joint structure of the description family and the target family. It does not require a unique winner on every target; ties may remain even when pairwise ordering is stable.

2.4 Stable winnerhood

Definition 2.2 (Stable winner). *A description $D^* \in \mathcal{D}$ is a stable winner over $\mathcal{T}$ within $\mathcal{D}$ if*

$$s(D^*; Y) \geq s(D_j; Y) \quad \text{for all } D_j \neq D^* \text{ and all } Y \in \mathcal{T}, \quad (2)$$

with strict inequality on at least one target for each competitor.

A stable winner is a description that is not dominated on any prediction target within the family. In the exact analytic setting, this requires simultaneous optimality. In the empirical export section (Section 6), stable winnerhood is operationalised by a pre-registered rule defined separately from the theorem; that operationalisation is intentionally distinct from the exact definition above.

3 Task-Relative Descriptive Privilege in Linear Gaussian Dynamical Systems

3.1 System and observer family

Let the state evolve as

$$x_{t+1} = Ax_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, Q), \quad x_t \in \mathbb{R}^n, \quad (3)$$

with A stable (spectral radius less than one) and $Q \succ 0$. The unique stationary covariance Σ satisfies $A\Sigma A^\top + Q = \Sigma$ and is strictly positive definite.

The description family is the rank- r linear observer class: each description is parametrised by a row-orthonormal matrix $U \in \mathbb{R}^{r \times n}$, $UU^\top = I_r$, acting as

$$D_U(x) = U\Sigma^{-1/2}x. \quad (4)$$

Pre-whitening by $\Sigma^{-1/2}$ places all observers on a common isotropic footing. The observer space is the Grassmannian $\text{Gr}(r, n)$.

The target family consists of squared-error prediction targets of the form $Y_{C,\tau} = Cx_{t+\tau}$, with $C \in \mathbb{R}^{p \times n}$ and integer $\tau \geq 1$. Performance is total mean squared prediction error.

3.2 Target information matrices and exact risk

Given observation $z = U\Sigma^{-1/2}x_t$, the MMSE predictor of $Cx_{t+\tau}$ is linear by Gaussian conjugacy. The Bayes risk is

$$R(U; C, \tau) = \text{tr}(C\Sigma C^\top) - \text{tr}(UM_{C,\tau}U^\top), \quad (5)$$

where the *target information matrix* is

$$M_{C,\tau} = \Sigma^{1/2}(A^\tau)^\top C^\top C A^\tau \Sigma^{1/2} \in \mathbb{R}^{n \times n}. \quad (6)$$

$M_{C,\tau}$ is symmetric positive semidefinite. The formula is exact; no approximation is involved.

Setting the score $s(D_U; Y_{C,\tau}) = -R(U; C, \tau)$ (higher score = lower risk), minimising risk over $\text{Gr}(r, n)$ is equivalent to maximising $\text{tr}(UM_{C,\tau}U^\top)$.

3.3 Top-eigenspace optimality

Proposition 3.1 (Ky Fan optimality). *For fixed rank r , the observer minimising $R(U; C, \tau)$ over $\text{Gr}(r, n)$ has row space equal to the top- r eigenspace of $M_{C,\tau}$. The minimum risk is*

$$R^* = \text{tr}(C\Sigma C^\top) - \sum_{i=1}^r \lambda_i(M_{C,\tau}), \quad (7)$$

where $\lambda_1 \geq \dots \geq \lambda_n$ are the eigenvalues of $M_{C,\tau}$. Uniqueness (up to rotation within the eigenspace) holds when $\lambda_r > \lambda_{r+1}$.

Proof. Minimising R is equivalent to maximising $\text{tr}(UM_{C,\tau}U^\top)$ subject to $UU^\top = I_r$. By the Ky Fan variational principle, the maximum equals $\sum_{i=1}^r \lambda_i(M_{C,\tau})$, achieved exactly when the row space of U spans the corresponding eigenspace. $\square$

3.4 Main theorem

The framework vocabulary of Section 2 maps onto the LGDS setting as follows. Each rank- r observer U defines a description D_U . The score is $s(D_U; Y_{C,\tau}) = -R(U; C, \tau)$. A description D_U is a stable winner over $\mathcal{T}$ within the rank- r class if its row space simultaneously minimises risk for every target in the family.

Definition 3.2 (r -coherence). *A target family $\mathcal{T} = \{(C_k, \tau_k)\}$ is r -coherent with respect to (A, Σ) if all information matrices $\{M_{C_k, \tau_k}\}$ share a common top- r eigenspace V^* .*

In LGDS, r -coherence is the structural condition that characterises when a stable winner exists within the rank- r observer class. The proof below is a direct consequence of the Ky Fan variational principle; the contribution is the packaging of coherence and stable winnerhood as named conditions with an empirical diagnostic that can be deployed outside the linear Gaussian class.

Theorem 3.3 (Stable winner iff r -coherent). *In a finite-dimensional LGDS at stationarity, among rank- r linear observers and squared-error prediction targets $\mathcal{T} = \{Y_{C_k, \tau_k}\}$:*

- (i) **Existence.** *A stable winner exists over $\mathcal{T}$ within the rank- r observer class if and only if $\mathcal{T}$ is r -coherent.*
- (ii) **No-global-privilege corollary.** *If two targets in $\mathcal{T}$ have information matrices with distinct top- r eigenspaces, no rank- r observer is simultaneously optimal for both; no stable winner exists.*
- (iii) **Coherent case.** *When $\mathcal{T}$ is r -coherent with common eigenspace V^* , the unique stable winner (up to rotation) is the observer whose row space equals V^* .*

Proof. (i), *necessity.* Suppose D_{U^*} is a stable winner. By Proposition 3.1, optimality for target Y_{C_k, τ_k} requires the row space of U^* to equal the top- r eigenspace of M_{C_k, τ_k} . Since D_{U^*} must be optimal for every target in $\mathcal{T}$, all information matrices share the common eigenspace $\text{rowspan}(U^*)$. Hence $\mathcal{T}$ is r -coherent.

(i), *sufficiency.* Suppose $\mathcal{T}$ is r -coherent with common eigenspace V^* . Let U^* be any row-orthonormal basis for V^* . By Proposition 3.1, D_{U^*} minimises risk simultaneously for every target in $\mathcal{T}$ and is therefore a stable winner.

(ii). If M_{C_1, τ_1} and M_{C_2, τ_2} have distinct top- r eigenspaces, no single row space can equal both. By Proposition 3.1, no rank- r observer is optimal for both targets simultaneously.

(iii). Follows directly from the sufficiency direction of (i) and the uniqueness clause of Proposition 3.1 when $\lambda_r > \lambda_{r+1}$ for each target's information matrix. $\square$

3.5 r -coherence is structurally demanding

r -coherence requires every target in the family to be a linear functional of the same r -dimensional sufficient statistic, across all prediction horizons. This demands strong alignment between the dynamics (A, Σ) and the task family $\mathcal{T}$. It is an algebraic equality condition, not an approximate or asymptotic one.

Outside r -coherence, there is no rank- r observer that is simultaneously optimal for all targets. The cost of forcing any fixed observer onto a non-native target is given exactly by equation (5): the global-privilege gap

$$\Gamma = \min_{U \in \text{Gr}(r, n)} \max_k [R(U; C_k, \tau_k) - R_k^*] > 0 \quad (8)$$

measures the irreducible worst-case regret under task conflict.

Table 1: Regret matrix, no-global-privilege instance ($n = 8$, $r = 2$, seed 100). All values from the exact analytic Bayes risk formula.

Observer	Regret on T_1	Regret on T_2
U_1^* (optimal for T_1)	0.000	0.036
U_2^* (optimal for T_2)	0.763	0.000
U_{joint} (minimax)	0.026	0.026

3.6 Numerical illustration

We illustrate Theorem 3.3 with a concrete instance ($n = 8$, $r = 2$, seed 100).¹ Two targets $T_1 = ([e_1^\top; e_2^\top], 1)$ and $T_2 = ([e_5^\top; e_6^\top], 5)$ have information matrices whose top-2 eigenspaces subtend principal angles of 54.1° and 24.0° : no single rank-2 observer is simultaneously optimal. The minimax observer U_{joint} , found by projected optimisation over $\text{Gr}(2, 8)$, achieves $\Gamma = 0.026 > 0$ (Table 1).

A structurally coherent instance is constructed by design (normal A_{sym} , targets aligned to the common top-2 eigenspace): the maximum pairwise principal angle across all four targets is 0.000° to machine precision, and perturbations of magnitude $\varepsilon = 0.1$ to C (20 random draws) produce a maximum principal angle of 0.102 rad — well below the fragility threshold of 0.15 rad. Structural coherence is perturbation-robust, distinguishing it from accidental near-coincidences that break under small perturbations.

4 The Exported Diagnostic

4.1 From eigenspaces to sign flips

Theorem 3.3 is exact within LGDS. This section extracts from it an *empirical diagnostic* that can be evaluated in systems that are not LGDS.

The theorem’s operative condition — whether information matrices share a common top- r eigenspace — is not directly testable in systems where the dynamics and noise structure are unknown or nonlinear. What *is* testable in any system is the sign of pairwise score differences across targets.

The diagnostic rests on the following observation. In LGDS, when two targets have distinct top- r eigenspaces, no stable winner exists (Theorem 3.3(ii)). By the exact risk formula (5), the optimal observer for target k incurs strictly positive regret on any other target with a distinct eigenspace. This eigenspace separation implies that no single observer is simultaneously optimal for both targets. In finite observer families, this loss of common optimality is naturally probed by pairwise score-gap sign changes across targets.

Conversely, if the target family is r -coherent, the same observer is optimal for every target in the rank- r class; this is the exact LGDS case in which no task conflict remains.

4.2 The exported prediction

This motivates a diagnostic that travels outside LGDS:

¹This numerical illustration recapitulates the construction from Bhatia [2026e] for self-containedness.

If at least one pairwise sign flip is empirically supported across a target family, no stable family winner should exist.

This is a prediction, not a theorem. Outside LGDS, sign flips are sufficient evidence of incoherence in the general sense of Section 2, but the connection to stable winnerhood is no longer guaranteed by Theorem 3.3. The export test asks whether the diagnostic nevertheless has empirical bite.

4.3 Operationalisation

To test the exported prediction, we require operational definitions of sign-flip support and stable winnerhood that are independent of any parametric model.

Supported sign flip. For an ordered pair of descriptions (D_i, D_j) and an estimable target $Y \in \mathcal{T}$, the gap $\Delta_{ij}(Y)$ is *supported in direction +* if its 95% bootstrap confidence interval excludes zero from below ($lo > 0$), and *supported in direction -* if the interval excludes zero from above ($hi < 0$). A pair (D_i, D_j) exhibits a *supported sign flip* across $\mathcal{T}$ if there exist two estimable targets Y_a, Y_b with opposite supported directions. A target family is *empirically incoherent* if at least one pair exhibits a supported sign flip; *empirically coherent* if every pair is sign-stable (all supported gaps point the same direction); and *unresolved* otherwise.

Stable family winner (empirical). Description D^* is a *stable family winner* if, for every competitor $D_j \neq D^*$:

- (i) D^* beats D_j on at least 75% of estimable targets with CI-supported gaps;
- (ii) no competitor beats D^* on more than 25% of estimable targets with CI-supported gaps.

Otherwise the cell is *winner-absent*.

These thresholds were fixed before examination of the regime-family results.

Correspondence test. The exported diagnostic is evaluated by the *correspondence test*: across pre-specified regime-family cells, does empirical incoherence predict winner-absence, and does empirical coherence predict stable winnerhood? The primary exported claim is the negative half: incoherence $\rightarrow$ winner-absent. The positive half (coherence $\rightarrow$ stable winner exists) requires coherent cells to be present in the battery; their presence or absence is itself a data outcome.

5 Relation to Prior Empirical Results

The empirical results motivating this paper were obtained before the formal framework above was developed. We briefly reposition them here as instances of coherence failure, not as evidence for a stronger unified law.

Cellular automata (P1, P4). In post-transient Conway’s Game of Life, a fine component-count observer and a coarse occupied-block observer disagree on net change over the same run. The disagreement is predictable from early fine-scale structure ($r = -0.848$, 95% CI $[-0.862, -0.830]$), and the scale that best predicts a future observable depends on which observable is chosen: future live-cell count peaks at block size $B = 2$, while future occupied-block count peaks at $B = 8$, with non-overlapping bootstrap confidence intervals [Bhatia, 2026b]. In the framework of Section 2, the target family $\{\bar{N}_{\text{live}}, \bar{N}_{\text{occ}}^{B=8}\}$ is empirically incoherent: the pairwise ordering between fine and coarse descriptions flips across the two targets. A stable winner is not observed over this family; the present theorem provides the formal vocabulary for understanding that kind of failure in the exact LGDS setting.

Kinetic Ising model (P2). Fine descriptions based on locally exposed defects outperform a coarse block-variance description for predicting future domain-wall change ($\Delta R^2 = +0.415$, 95% CI $[+0.389, +0.443]$) in the ordered phase, but the advantage reverses in the disordered phase ($\Delta R^2 = -0.064$) [Bhatia, 2026d]. The regime dependence shows that no regime-independent stable winner exists over the combined family of ordered and disordered targets.

Bose–Hubbard chain (P3). In an open Bose–Hubbard chain with Lindblad dephasing, targeted additional dephasing at high-variance sites produces more local occupation loss than matched-budget random targeting in the $J/U = 0.30$ – 0.40 regime, but the effect reverses at $J/U = 0.12$ [Bhatia, 2026a]. This is a causal instance of target-relative privilege. Because the causal axis lies outside the scope of the present theorem, we cite it here only as empirical pressure for a general winner-instability question, not as a theorem-governed instance.

N-body dynamics (P5). In three-dimensional self-gravitating N-body simulations, the coarse density variance robustly predicts future clustering for bimodal initial conditions ($r = -0.975$, 95% CI $[-0.99, -0.95]$), but the kinematic fine observable (local velocity dispersion) shows a suggestive advantage for concentrated profiles at small softening [Bhatia, 2026f]. The results suggest that winner identity may depend on initial-condition family, though the fine-side concentrated-profile signal remains unresolved. We cite this paper as pressure for the formal question, not as a settled theorem instance.

These four cases are not re-proved here. They are cited as recurring empirical pressure for the formal question the theorem addresses: when is a stable winner over a target family even possible? The theorem gives the answer for LGDS; the export test in Section 6 probes whether the answer generalises. Section 5 is deliberately subordinate to the theorem and export core; its purpose is motivation, not reinterpretation.

6 Evolution Export Test

6.1 Setup

We test the exported diagnostic in an individual-based evolutionary simulation. Four descriptions are compared: gene-frequency composition (D_{gene}), lineage-level structure (D_{lin}), organism-level state (D_{org}), and an organism-plus-ecology composite (D_{eco}). The target family consists of four evolutionary outcomes: persistence, transmission, dominance,

and ecological state. Two task families are evaluated separately: predictive (5-fold cross-validated R^2) and causal (mean outcome difference under matched interventions).

Six evolutionary regimes are tested: selected Wright-Fisher (**selected_wf**), Moran (**moran**), frequency-dependent selection (**freq_dep**), eco-evolutionary coupling (**eco_evol**), group-structured (**group_structured**), and neutral Wright-Fisher (**neutral_wf**). This yields 12 regime-family cells.

Full simulation parameters and the pre-registered battery design are reported in Bhatia [2026c].

6.2 Coherence and winner classification

For each regime-family cell, pairwise gaps $\Delta_{ij}(Y)$ are computed from pre-aggregated scores using the independence approximation:

$$\widehat{se}_{ij} = \sqrt{\widehat{se}_i^2 + \widehat{se}_j^2}, \quad (9)$$

where each $\widehat{se}_k$ is derived from the individual 95% bootstrap confidence interval as $\widehat{se}_k = (\text{ci_hi}_k - \text{ci_lo}_k)/(2z_{0.975})$. With $n = 100,000$ replicates per cell, the resulting intervals are narrow; however, the deviation from joint replicate-level bootstrapping remains a documented departure from the pre-specified protocol.

Coherence and winner labels are assigned by the rules in Section 4.²

6.3 Results

Table 2 reports coherence and winner labels for all 12 regime-family cells. In this pre-aggregated adapter analysis, eleven cells received a non-unresolved coherence label; one cell (**selected_wf/causal**) was unresolved and excluded from the correspondence test per the pre-registered rule.

Of the 11 classified cells:

- all 11 are empirically incoherent;
- all 11 are winner-absent;
- the correspondence test yields 0 mismatches.

Note on winner classification. In the current results file, organism-plus-ecology data is present only for the **eco_evol/predictive** cell; all other cells lack the fourth description entirely. Because the winner rule requires all four descriptions to be estimable, winner-absent in 10 of 11 classified cells is a default rather than a competitive outcome. The exception is **eco_evol/predictive**, where a competitive check ran across the four descriptions and found no winner. The incoherence result is unaffected: sign flips are established from the three descriptions that are present across all regimes.

Given this coverage gap, the adapter run is directionally consistent with the winner-absence prediction on the covered cells, but does not constitute a strong transport test. The

²The two classifiers consume a shared, deterministically seeded gap table; neither classifier re-randomises the same statistic, preventing Monte Carlo noise from producing artificial classification discrepancies.

Table 2: Coherence and winner classification for all 12 regime-family cells. Classified cells: 11. Unresolved: 1. Mismatches: 0.

Regime	Family	Coherence	Winner	Match
<code>selected_wf</code>	predictive	incoherent	winner-absent	✓
<code>selected_wf</code>	causal	unresolved	(excluded)	—
<code>moran</code>	predictive	incoherent	winner-absent	✓
<code>moran</code>	causal	incoherent	winner-absent	✓
<code>freq_dep</code>	predictive	incoherent	winner-absent	✓
<code>freq_dep</code>	causal	incoherent	winner-absent	✓
<code>eco_evol</code>	predictive	incoherent	winner-absent	✓
<code>eco_evol</code>	causal	incoherent	winner-absent	✓
<code>group_structured</code>	predictive	incoherent	winner-absent	✓
<code>group_structured</code>	causal	incoherent	winner-absent	✓
<code>neutral_wf</code>	predictive	incoherent	winner-absent	✓
<code>neutral_wf</code>	causal	incoherent	winner-absent	✓

correspondence test returns a formal pass under the locked classification rule; that verdict should be read as adapter-consistent rather than strongly confirmed.

The dominant sign flip is the pair $(D_{\text{gene}}, D_{\text{lin}})$, which appears in 9 of 11 classified cells across both task families. This pair flips direction across the persistence/transmission divide: gene-frequency descriptions are more predictive of transmission outcomes in several regimes, while lineage descriptions are more predictive of persistence outcomes, with bootstrap-supported gaps in both directions.

No coherent cells were observed in this battery. The positive half of the exported prediction — that coherence should imply stable winnerhood — therefore remains untested.

6.4 Data coverage note

After correcting the `org_eco` $\rightarrow$ `organism_ecology` name mapping, organism-plus-ecology data is present in the results file only for the `eco_evol`/predictive cell; all other regimes and the full causal family have no organism-plus-ecology data. Because the pre-registered winner rule requires all four descriptions to be estimable on a target before that target counts toward the winner criterion, winner classification is competitive only for `eco_evol`/predictive (where a competitive check ran and found no winner) and vacuous (returning winner-absent by default) for all remaining cells. The incoherence classification is unaffected: the $(D_{\text{gene}}, D_{\text{lin}})$ sign flip is established from the three descriptions that are present across all regimes. A full four-description rerun is required before winner classification can be treated as a genuine competitive result across the battery.

6.5 What is and is not claimed

The evolution export test supports the following claim:

In the pre-aggregated adapter analysis of the tested evolutionary simulation

battery, the coherence diagnostic derived from Theorem 3.3 correctly predicts winner-absence in all classified incoherent regime-family cells (11 of 11; 0 mismatches).

It does not support:

- that the LGDS theorem is proved outside LGDS;
- that evolutionary systems are approximately linear-Gaussian;
- that the positive half of the diagnostic (coherence $\rightarrow$ stable winner) holds in this system, since no coherent cells were observed;
- that winner-absence has been competitively established across all four descriptions in all classified cells — only `eco_evol/predictive` ran a full four-description competitive check; all other winner-absent labels are defaults due to missing organism-plus-ecology data;
- that the result generalises beyond the six tested regimes or the four tested descriptions.

7 Scope and Limitations

Exactness of the theorem. Theorem 3.3 is exact within finite-dimensional linear Gaussian dynamical systems at stationarity, among rank- r linear observers and squared-error prediction targets of the form $Y = Cx_{t+\tau}$. The proof is algebraic and involves no approximation. The scope is narrow by design: the theorem is a precise answer to a precise question, not a universal law about description-dependent privilege.

What the theorem does not cover. The result does not apply to nonlinear systems, non-Gaussian noise, nonlinear observers, or prediction tasks with non-quadratic loss. The causal axis — whether the observer optimal for prediction differs from the observer optimal for intervention — is not addressed by the theorem; it requires a separate formal treatment. Temporal asymmetry and the direction of optimal description are also outside the theorem’s scope.

Status of the export test. The evolution export test (Section 6) asks whether the coherence diagnostic has empirical bite outside LGDS. Current support is strongest for the negative half: all 11 classified incoherent cells are winner-absent (0 mismatches). The positive half — that coherence should imply stable winnerhood — remains untested because no coherent cells appeared in the current battery. The diagnostic travels in the tested direction; the full bidirectional claim is not yet earned.

The adapter result is subject to two documented limitations. First, pairwise gap CIs are computed via the independence approximation from pre-aggregated scores rather than joint replicate-level bootstrapping as pre-specified; this is a registered departure. Second, the current adapter result is limited by incomplete four-description coverage in the results file. After correcting the `org_eco` name mapping, organism-plus-ecology data is present only

for the `eco_evol`/predictive cell; all other regimes and the full causal family lack that fourth description entirely. The residual `ecology` entry in `interv_class` is a separate column-leak issue. A full four-description rerun is required before winner classification can be treated as a genuine competitive result across the battery. Neither limitation currently appears likely to change the qualitative verdict, but both must be resolved before the result is treated as final.

Generality of the diagnostic. The sign-flip diagnostic is motivated by the LGDS theorem but is not proved to characterise stable winnerhood in any other system class. Its application to evolutionary simulations is an empirical test of transportability, not a theorem extension. Positive results in further system classes would strengthen the case for the diagnostic as a general tool; null results would narrow its scope. The program of mapping which system classes admit or fail the diagnostic is open.

Relation to prior formalisms. The framework in Section 2 is related to, but distinct from, several prior formalisms. The renormalisation group and the Wilson–Kadanoff coarse-graining program ask which features are preserved under scale transformations; the present framework asks which description is optimal for a given prediction task, a question that is orthogonal to preservation under transformation. Hoel et al.’s causal emergence asks when macro descriptions carry more effective information than micro descriptions; the present result is complementary but narrower, characterising the structural conditions under which a single description is globally privileged across a task family rather than whether macro beats micro on a fixed task. Israeli and Goldenfeld’s computational irreducibility result establishes that some cellular automata resist exact coarse-graining [Israeli and Goldenfeld, 2004]; the incommensurability of the two observers in the CA results is consistent with that finding.

8 Discussion

What changes in how we read the prior results. The four prior empirical results cited in Section 5 each reported that winner identity depended on the target or the regime. Without a formal criterion, these findings could be read as isolated system-specific curiosities rather than instances of a structural phenomenon. The theorem sharpens that reading — not by proving those results are LGDS instances, but by supplying the vocabulary and the criterion that make the pattern recognisable. A result in which fine-description advantage reverses between phases is now legible as a candidate instance of coherence failure over a joint regime-indexed target family: the same description does not remain best once the target family is enlarged across regimes. The theorem does not prove that these systems instantiate the LGDS mechanism; it provides the formal vocabulary for stating the winner-stability question precisely and, in the LGDS class, answering it exactly.

Open problems. Four problems remain open after this paper.

First, the positive half of the exported diagnostic is untested. The evolution battery produced no coherent regime-family cells, so the prediction that coherence implies stable

winnerhood was never challenged. A battery designed to include coherent cells — either by construction in a synthetic system or by identifying regimes where one description is known to dominate — is needed before the bidirectional claim can be assessed.

Second, the causal axis remains untreated. The theorem concerns prediction under the squared-error criterion; it says nothing about which description is optimal for intervention. The prior results include causal instances (Bose–Hubbard, evolution) where the best predictive description and the best causal handle differ. A formal treatment of causal-axis coherence, if it exists, would require a different object — likely a response-theoretic functional rather than an information matrix.

Third, the diagnostic’s scope outside LGDS is unknown. The evolution export test supports the negative half in one system class. Whether the same diagnostic works in continuous-field systems, in systems with non-quadratic loss, or in systems with non-stationary dynamics is an open empirical question. Null results in further system classes would be as informative as positive ones: they would identify the boundary conditions under which coherence structure predicts winner-stability.

Fourth, a formal generalisation of the theorem is open. The LGDS result is exact because the Bayes risk formula is exact and the Ky Fan variational principle applies directly. Analogous exact results in nonlinear or non-Gaussian systems would require either a different risk characterisation or an approximate treatment. Whether the coherence condition has a natural extension beyond the linear Gaussian class — and whether that extension recovers the LGDS theorem as a special case — is the formal problem the current result points toward.

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